Methods for electrochemical reaction sampling with machine-learned force fields at AIMD-level accuracy

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Abstract

Groundwork for ab initio molecular dynamics (AIMD)–accurate sampling of electrochemical interfaces using machine learning force fields (MLFFs): arbitrary electrolyte mixtures, stability of MLFF molecular dynamics, and reproducible training-data generation.

1 Introduction

2 Background

3 Methods

3.1 Training-data generation

3.2 Mixture systems and composition dependence

3.3 MLFF molecular dynamics and stability criteria

4 Results

5 Discussion

6 Conclusion

Data and software availability

Tutorial and library dependencies: https://github.com/cap-jmk-real/mlff-tutorial, https://github.com/cap-jmk-real/mond, and https://github.com/cap-jmk-real/openmm-mixtures. Research artifacts: https://github.com/cap-jmk-real/aimd-mlff-methods (code MIT, paper text CC BY 4.0).